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A Lossless Compression

Compressing, transmitting and archiving important medical reports is just what the doctor needed

Mithun Kidambi

Ever wondered how hundreds of echocardiograms get stored in a hospital? Actually... how many of us even know what an echocardiogram is? Simply put, an echocardiogram is a study of the heart using high-frequency sound waves.

The objective of an echocardiogram is to visualise the heart chambers, thickness of the muscle wall, heart valves and major blood vessels located near the heart. A non-invasive procedure, there is no cutting and sewing involved. The picture obtained is more detailed than an X-Ray image. This test is used in the diagnosis of a number of heart conditions.

Now that you are familiar with an echocardiogram, let us understand the importance of image and data compression technology to store multiple echocardiogram records without any quality loss.

Archiving, A Necessity

A lot of hospitals and clinics around the world keep an archive of these records in an analogue format, such as cassette tapes. In the West, the law makes it necessary for clinics and hospitals to maintain these records for a period of at least seven years, whereas India has no such law pertaining to medical records.

In the event of such legislation here too, hospitals will have to invest lots of money into setting up and maintaining databases, analogue or oth-



Illustration Akhilesh Prasad